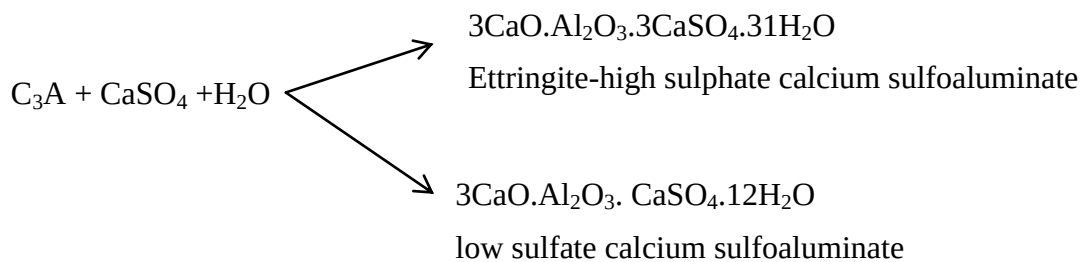


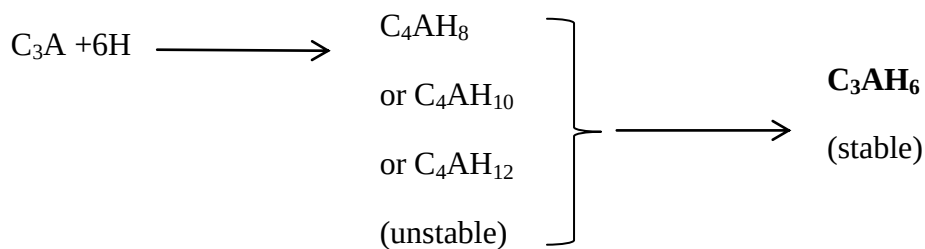
2- Tricalcium aluminate hydrate and the action of gypsum:

The amount of C_3A present in most cements is comparatively small but its behaviour and structure relationship with other phases in cement make it of interest. C_3A acts as a flux and thus reduces the temperature of burning of clinker and facilitates the combination of lime and silica. However, C_3A contributes little or nothing to the strength of cement except at early ages when hardened cement paste is attacked by sulfates, expansion due to the formation of calcium sulfoaluminate from C_3A may results in a disruption of the hardened paste.

The reaction of pure C_3A with water is very violent and leads to immediate stiffening of the paste, known as “flash set”. To prevent this from happening, gypsum ($CaSO_4 \cdot 2H_2O$) is added to cement clinker. Gypsum and C_3A react to form insoluble calcium sulfoaluminate ($3CaO \cdot Al_2O_3 \cdot 3CaSO_4 \cdot 31H_2O$), but eventually tricalcium aluminate hydrate is formed, although this is present by a metastable ($3CaO \cdot Al_2O_3 \cdot CaSO_4 \cdot 12H_2O$), produced at the expense of the original high-sulfate calcium sulfoaluminate.



As more C_3A comes into solution, the composition changes, the sulphate content decreasing continuously. The rate of reaction of the aluminate is high and, if this readjustment in composition is not rapid enough, direct hydration of C_3A is likely.



Therefore there is an **optimum gypsum content (O.G.C)** which is the amount of gypsum to be added to the cement and results in: high strength, low shrinkage after drying and does not cause large expansion in the concrete during the curing:

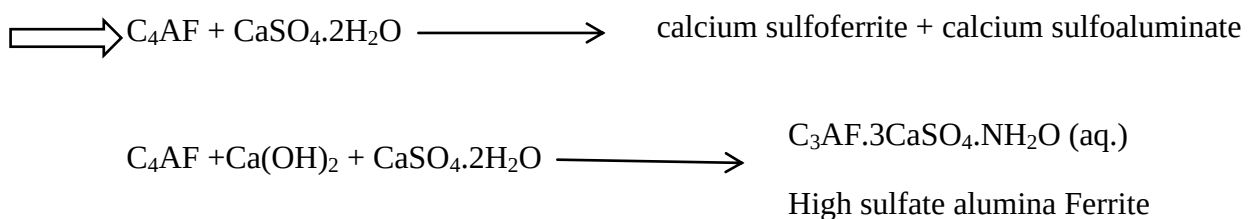
- The amount of O.G.C **increases** with the increase in: C_3A content, the alkali content of the cement, fineness of cement, curing temperature and time of curing.
- The amount of O.G.C **decreases** with the increase in the percentage of free CaO , free MgO and the chlorides , which could be in the aggregate, or water, or when added as accelerated materials to the cement.

Notes:

- ✓ The amount of gypsum added to the cement clinker has to be carefully watched; in particular an excess of gypsum leads to an expansion and consequent disruption of the set of cement paste.
- ✓ The amount of gypsum added to cement clinker is expressed as the mass of SO_3 present; this is limited by European Standard ENV 197-1:1992 to a maximum of 3.5%, but in some cases higher percentages are permitted.

3- Hydration of C_4AF :

- It is believed to hydrate into tricalcium aluminates hydrate and amorphous phase, probably $CaO.Fe_2O_3 \cdot aq$.
- It is possible that some Fe_2O_3 is present in solid solution in the tricalcium aluminate hydrate.



Setting:

Setting refers to a change from a liquid to a rigid state,

Cement + water → cement paste → lose its plasticity gradually → when it lose its plasticity completely → setting occurs.

➤ **The stages of setting include:**

- Initial setting
- Final setting

Initial setting time – refers to the beginning of the cement paste setting.

Final setting time – refers to the beginning of hardening and gain of strength.

Iraqi Standard Specification No. 5 limits:

- Initial setting time not less than 45 minutes
 - Final setting time not more than 10 hours.
- It is important to distinguish **setting** from **hardening**, which refers to the gain of strength of a set cement paste.
- The two first to react are C_3A and C_3S .
- The importance of setting in concrete works comes from the importance to keep the fresh concrete in the plastic stage for enough time necessary to complete its mixing and placing under practical conditions. But, from the economical side, it is important that the concrete hardens at convenient period after casting.

Factors affecting the setting

- 1- Water/cement (w/c) ratio** – The setting time of cement increases with the increase of w/c ratio.
- 2- Temperature and relative humidity** - The setting time of cement decreases with a rise in temperature and decrease of relative humidity.
- 3-Fineness of cement** - The setting time of cement decreases with a rise in fineness of cement

Flash setting :

Occurs when there is no gypsum added or exhausting the gypsum (added with little amount), so C_3A reacts violently with water causing liberation high amount of heat causing rapid setting of cement, and leading to form porous microstructure that the product of hydration of the other compounds precipitate through, unlike the **normal (ordinary) setting** that have much lower porosity microstructure.

False setting:

It is an abnormal premature stiffening of cement within a few minutes of mixing with water. It differs from “flash setting” in that :

- No appreciable heat is evolved,
- Remixing the cement paste without addition of water restores plasticity of the paste until it sets in the normal and without a loss of strength.

✓ Causes of false setting:

- 1- Dehydration of gypsum when interground with too hot a clinker: hemihydrate ($CaSO_4 \cdot \frac{1}{2}H_2O$) or anhydrite ($CaSO_4$) are formed and when cement is mixed with water these hydrate to form needle-shaped crystals of gypsum.
- 2- Bad storage: during storage the alkalis in the cement may carbonates, and alkali carbonate react with $Ca(OH)_2$ liberated by the hydrolysis of C_3S to form $CaCO_3$. This precipitates and induces a rigidity of the paste.
- 3- Activation of C_3S by aeration at moderately high humidities: water is absorbed on the grains of cement, and these freshly activated surfaces can combined very rapidly with more water during mixing: this rapid hydration would produce false set.

Fineness of cement:

Grinding the clinker to the required fineness is considered as the last step in the manufacture of Portland cement.

✓ *Advantages of high fineness of cements:*

- 1- The hydration starts at the surface of the particles of cement, it is the total surface area of cement that represents the material available for hydration. Thus, the rate of hydration depends on the fineness of cement particles. For a rapid development of strength, higher fineness is necessary (Figure 1.5), the long-term strength is not affected.
- 2- A higher early rate of hydration means a higher rate of early heat evolution
- 3- An increase in fineness of cement slightly improves the workability of a concrete mix.
- 4- Reduce bleeding in concrete.

✓ *Disadvantages of high fineness of cement:*

- 1- Increase the cost of grinding the clinker.
- 2- The finer the cement the more rapidly it deteriorates on exposure to the atmosphere.
- 3- Finer cement leads to a strong reaction with alkali-reactive aggregate and makes the cement paste, though not necessary concrete, exhibit at higher shrinkage and proneness to cracking.
- 4- An increase in fineness increases the amount of gypsum required for proper retardation because, in finer cement, more C_3A is available for early hydration.
- 5- The water content of a paste of standard consistency is greater the finer the cement.

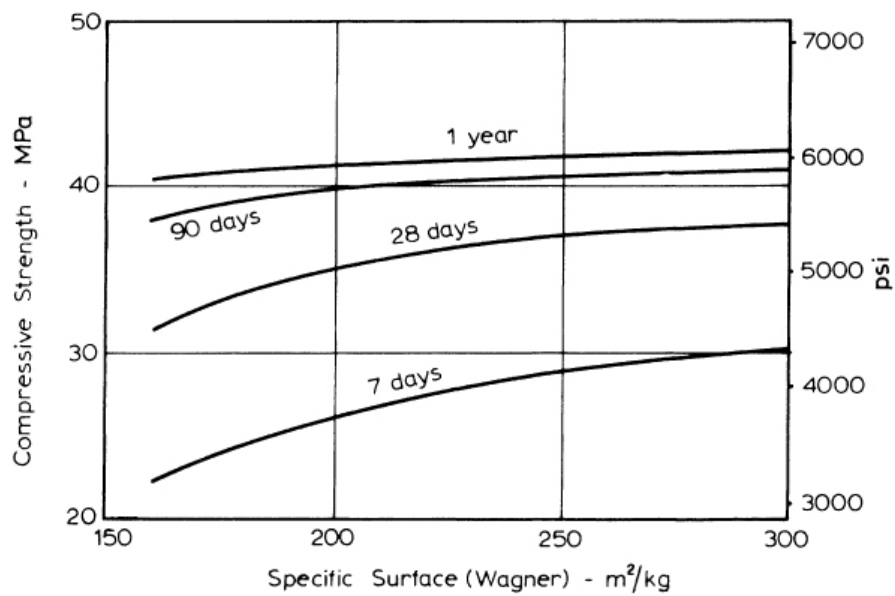


Fig. 1.5. Relation between strength of concrete at different ages and fineness of cement¹

Note: * fineness is determined by using sieve No. 325 (45 μm) according to ASTM C-403.

*Many methods are used to determine the fineness of cement such as:

- Wanger tubidimate.
- Air permeability developed by lea and nurse.
- Blain method.

Structure of hydrated cement:

The mechanical properties of hardened cement and concrete appear to depend not so much on the mechanical composition of the hydrated cement as on the physical structure of the products of hydration, viewed at the level of colloidal dimensions.

At any stage of hydration, the hardened paste consists of:

- 1- Very poorly crystallized hydrates of the various compounds, referred to collectively as gel, of crystals of Ca(OH)_2 .
- 2- Some minor compounds.
- 3- Unhydrated cement
- 4- The residual of the water-filled spaces in the fresh paste. These voids are called :capillary pores, but, within the gel itself, there exists interstitial voids, called gel pores

With the process of hydration:

- 1- Since the products of hydration are colloidal during hydration the surface area of the solid phase increases enormously.
- 2- Large amount of free water becomes absorbed by the products of hydration.
- 3- If no water movement to or from the cement paste is permitted, the reactions of hydration use up the water until too little is left to saturate the solid surfaces.
- 4- The relative humidity within the paste decreases which causes (self-desiccation).
- 5- Since gel can form only in water-filled space, self-desiccation leads to a lower hydration compared with a moist-cured paste.
- 6- Self-desiccation pastes with water/cement ratios in excess of 0.5, the amount of mixing water is sufficient for hydration to process at the same rate as when moist-cured.

✓ **Capillary pores:** at any stage of hydration, the capillary pores represent that part of the gross volume which has not been filled by the products of hydration.

- Since the products of hydration occupy more than twice the volume of the original solid phase (i.e cement) alone, the volume of the capillary system is reduced with the progress of hydration.
- The capillary porosity of the paste depends both on:
 - 1- w/c ratio of the mix.
 - 2- Degree of hydration.

Properties of capillary pores:

1- Capillary pores cannot be viewed directly but their median size was estimated from vapour pressure measurements to be about 1.3 μm .

2- They vary in shape, but shown by measurements of permeability, form an interconnected system randomly distributed throughout the cement paste.

3- The interconnected capillary pores are mainly responsible for the permeability of the hardened cement paste and for its vulnerability to cycles of freezing and thawing.

- Hydration increases the solid content of the paste and, in mature and dense pastes, the capillaries can become blocked by gel and segmented so that they turn into capillary pores interconnected solely by the gel pores.
- The absence of continuous capillaries is due to a combination of:
 - 1- A suitable w/c ratio: if w/c increases, the interconnected capillaries increases but if w/c ratios above about 0.7, even complete hydration would not produce enough gel to block all the capillaries.
 - 2- Sufficient long period of moist curing.

- ✓ **Gel pores:** These are interconnected interstitial spaces between the gel particles, which are needle-plate-, and foil-shaped.

Properties of gel pores:

- 1- The gel pores are much smaller than the capillary pores less than 2 or 3 nm in nominal diameter.
- 2- The gel pores occupy about 28% of the total volume of gel. The actual value is characteristics for a given cement but is largely independent of the w/c ratio of the mix and of the progress of hydration.

This indicate that:

- 1- Gel of similar properties is formed at all stages.
- 2- Continued hydration does not affect the products already in existence.
- 3- As the total volume of gel increases with the progress of hydration, the total volume of gel pores also increases but the volume of capillary pores decreases with the progress of hydration.

Volume of products of hydration:

The gross space available for the products of hydration consists of : the absolute volume of the dry cement + the volume of water added to the mix.

Total volume of gel (V_g) = Volume of solid products ($V_{s,p}$) + Volume of gel water (V_{gw})

Note:

- In this stage, the small loss of water due to bleeding and contraction of the paste while still plastic will be ignored.
- The water bond chemically by C_3S and C_2S was shown to be very approximately 24 % and 21 % of the mass of the two silicates, respectively.

C_3S = 24 % by weight

C_2S = 21 % by weight

C_3A = 40% by weight

C_4AF = 37 % by weight

The numbers above are not accurate as no enough information is available to determine the amount of the chemically combined water with the cement. Therefore, it is consider the volume of (non-evaporated water), known as volume of combined water) to be equal to (0.23 × volume of absolute volume of cement):

$$V_{S,P} = V_C + V_{cw} (1-0.254)$$

$$V_{S,P} = V_C + 0.23 W_c (1-0.254)$$

Where:

V_{cw} :volume of combined water.

W_c = weight of water.

V_c = absolute volume of cement

$V_{S,P}$ = volume of solid products